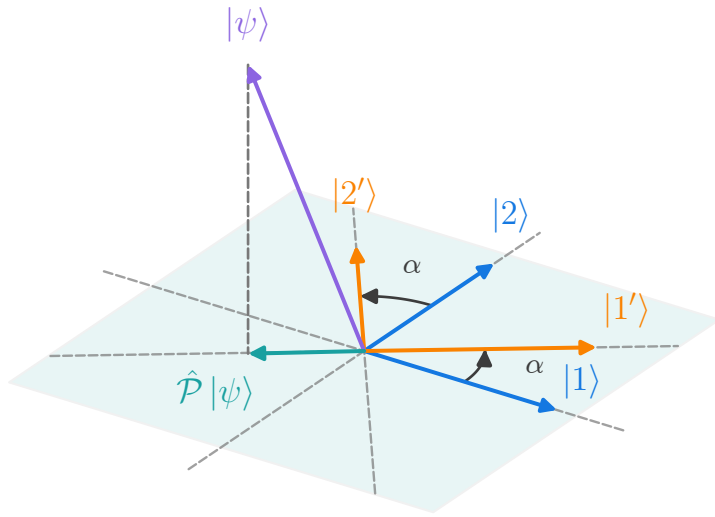




different bases in one subspace

$$|1'\rangle = \cos \alpha |1\rangle + \sin \alpha |2\rangle$$

$$|2'\rangle = -\sin \alpha |1\rangle + \cos \alpha |2\rangle$$

$$\alpha = \pi/5$$

$$\hat{\mathcal{P}}' = \hat{\mathcal{P}}$$